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ORIGINAL RESEARCH

Cardiology

Wearable devices for out-of-hospital cardiac arrest: A population survey on the willingness to adhere

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Abstract

Objectives: When an out-of-hospital cardiac arrest (OHCA) occurs, the first step in the chain of survival is detection. However, 75% of OHCAs are unwitnessed, representing the largest barrier to activating the chain of survival. Wearable devices have the potential to be "artificial bystanders," detecting OHCA and alerting 9-1-1. We sought to understand factors impacting users' willingness for continuous use of a wearable device through an online survey to inform future use of these systems for automated OHCA detection.

Methods: Data were collected from October 2022 to June 2023 through voluntary response sampling. The survey investigated user convenience and perception of urgency to understand design preferences and willingness to adhere to continuous wearable use across different hypothetical risk levels. Associations between categorical variables and willingness were evaluated through nonparametric tests. Logistic models were fit to evaluate the association between continuous variables and willingness at different hypothetical risk levels.

Results: The survey was completed by 359 participants. Participants preferred hand-based devices (wristbands: 87%, watches: 86%, rings: 62%) and prioritized comfort (94%), cost (83%), and size (72%). Participants were more willing to adhere at higher levels of hypothetical risk. At the baseline risk of 0.1%, older individuals with prior wearable use were most willing to adhere to continuous wearable use.

Conclusion: Individuals were willing to continuously wear wearable devices for OHCA detection, especially at increased hypothetical risk of OHCA. Optimizing willingness is not just a matter of adjusting for user preferences, but also increasing perception of urgency through awareness and education about OHCA.

KEYWORDS
adherence, OHCA, out-of-hospital cardiac arrest, wearable devices

1 | INTRODUCTION

1.1 | Background

Out-of-hospital cardiac arrest (OHCA) is a major health issue, with a global average incidence of 111 adults per 100,000 person-years (~40 million).[1] When an OHCA occurs, detection is paramount. This is followed by the initiation of cardiopulmonary resuscitation (CPR) and rapid defibrillation (automated external defibrillator -AED) ideally prior to the arrival of emergency medical services (EMS). In unwitnessed OHCAs, detection is the largest barrier to the activation of the chain of survival and notification of EMS.[2–5] Detection relies on the presence of a bystander, but approximately 75% of OHCAs occur in isolation.[6] This delays intervention, with the odds of survival dropping by 13% for every 1-min delay in recognition[7]; as such, approximately 50% of unwitnessed cases receive no treatment at all.[8] Even for unwitnessed cases for whom EMS elects to attempt resuscitation, only 2%–4% survive to hospital discharge.[9]

OHCA causes a cascade of physiological changes. Notably, cardiac output and body motion cease, respiratory rate may slow and become irregular (eg, agonal breathing), and dangerous cardiac rhythms (eg, ventricular fibrillation and asystole) can often be detected.[10–12] Additionally, significant reductions in blood oxygenation are observed and blood pressure drops to zero.[11,13] Characterizing OHCA as a combination of these detectable physiological parameters can inform the development of timely detection strategies. Many wearable devices utilize sensors, such as electrocardiography (ECG), photoplethysmography (PPG), and inertial measurement units (IMUs), that can continuously monitor these physiological parameters and their changes associated with OHCA.[14,15]

1.2 | Importance

Recently, wearable devices have gained popularity within the sports performance (eg, Apple Watch and FitBit)[16,17] and clinical sectors (eg, ZioPatch and Owlet SmartSock),[18,19] with over 400 million connected wearable devices in North America.[20] Such devices offer users direct access to physiological data (eg, heart rate) and allow healthcare professionals to monitor patient health, making them attractive options for both users and healthcare professionals. Additionally, the variety in form factors (eg, watch and chest strap) makes wearable devices more customizable and convenient for users. Despite the increasing adoption of wearables for health monitoring, there are currently no commercial wearables designed to detect OHCA.[14] Additionally, user adherence to continuously wear a device for OHCA monitoring presents a barrier to widespread adoption.

In the context of medical wearables, we suggest that user adherence can be broadly understood as a function of (1) user convenience (eg, form factor, usability, and accessibility) and (2) user perception of urgency (eg, personal/family medical history and knowledge of OHCA). Understanding how user convenience and perception of urgency affect adherence is critical to designing a wearable device for OHCA detection and encouraging its broad adoption.

The Bottom Line

75% of out-of-hospital cardiac arrests (OHCA) are completely unwitnessed, with the chances of survival diminishing if it is not identified within 20 minutes. Wearable sensors have the potential to act as artificial bystanders, witnessing unwitnessed events, and improving survival. However, are people willing to continuously wear a sensor for OHCA detection? If so, what form factor would they prefer to wear? We conducted a pan-Canadian survey to better understand the factors impacting an individual's decision to use a wearable continuously. We highlighted recommendations for the widespread implementation of such devices to aid developers in future design and improve OHCA detection overall.

1.3 | Goals of this investigation

There have been surveys within the field of wearables primarily investigating the efficacy of specific interventions (eg, the use of wearables amongst individuals with mental health diagnoses),[21,22] opinions of specific groups on wearable use (eg, healthcare workers),[23] or uptake of wearables within specific populations (eg, older adults).[24] In this study, we sought to identify how user convenience and perception of urgency influence wearable adherence in the general public.

2 | METHODS

This study employed a pan-Canadian online survey engaging with a hypothetical wearable device for OHCA detection. As such, the survey measured participants' willingness to adhere to continuous wearable use, as opposed to adherence itself.

2.1 | Survey development and distribution

This study was approved by the university Research Ethics Board (H21-03589) and used the online Qualtrics Survey Tool.[25] We created the survey (Supporting Information Appendix A) through examining previous validated surveys on adherence to medications (eg, Morisky Medication Adherence Scale) and consulting with 11 individuals (OHCA experts, users with lived OHCA experience, and users with no prior OHCA association).[26] We piloted the survey to evaluate question coherence and comprehensiveness.

The survey was publicly available from October 2022 to June 2023. Advertising was done through posters (eg, bulletin boards and mailing lists), social media (eg, LinkedIn and X), outreach to target populations (eg, older individuals with prior cardiorespiratory conditions), and an online research participation platform in British Columbia, Canada (ReachBC).[27] A voluntary response sampling approach was utilized in distribution, with the survey topic included in outreach materials. Prior to taking the survey, participants provided informed consent through an online consent form.

2.2 | Survey structure

The survey questionnaire was divided into three blocks: (1) background questions, (2) scenario-specific questions, and (3) demographic questions.

2.2.1 | Background questions

This block investigated user convenience in terms of form factors (eg, watch, ring, and chest strap) and factors impacting decisions to use a wearable (eg, cost, size, and battery life). This block also classified participants according to two statuses: (1) *current/past use of wearable devices (user status)*: whether participants previously used a wearable device; and (2) *awareness of cardiac arrest (awareness status)*: whether participants had a personal/family history of cardiopulmonary health conditions, or whether they knew someone who experienced a cardiac arrest (Supporting Information).

2.2.2 | Scenario-specific questions

This block investigated the impact of user perception of urgency on their willingness to adhere to continuous wearable use. Willingness to adhere was assessed in relation to different levels of hypothetical risk of experiencing OHCA, including (1) a baseline risk of 0.1%; (2) an increased risk of 2% (IR 1); and (3) an increased risk of 10% (IR 2). The increased hypothetical risk levels were intended to vary participants' perception of urgency and were presented through fractions and visual representations (Supporting Information Appendix A). At each risk level, participants were asked to rank their willingness (0%–100%) to use a wearable (1) once, (2) continuously, and (3) during specific activities (eg, physical exercise and sleep).

2.2.3 | Demographic questions

This block collected age, gender, sex, geographic location, geographic population, academic level, and profession. Biological sex is more likely to inform on physiological risk for OHCA, wherein males are more likely to experience a cardiac arrest.[1] However, we speculate that gender identity is more relevant for design preferences.

2.3 | Data analysis

Analysis was conducted using the R statistical package (Foundation for Statistical Computing, Vienna, version 4.2.2). Participants were grouped under four age categories: (1) under 25, (2) 25–45, (3) 45–65, and (4) over 65 to describe age-dependent preferences. A survey response was deemed incomplete if the first two blocks were incomplete. Survey responses with incomplete demographic data were only included in the analysis of general trends.

2.3.1 | User convenience

To assess user convenience, we identified the percentage of participants that preferred specific form and usability/accessibility factors, disaggregated by demographics and status-groups. Participants were able to select more than one preference in form factors and usability/accessibility factors.

2.3.2 | Perception of urgency

Next, we quantified user perception of urgency by comparing willingness to adhere at the three hypothetical risk levels. This was evaluated over the full cohort, and within demographic (age and gender) and status (user and awareness) subgroup divisions.

Prior to further analysis, willingness and age data were assessed for normality using a Shapiro–Wilk test and it was revealed that the data were not normally distributed. We thus tested associations between demographic/status groups and willingness using the Wilcoxon rank-sum test (for two-group comparisons) or the Kruskal–Wallis test (for $\geq$2-group comparisons) over and within the three hypothetical risk levels. Within age, we applied logistic regression within each of the three risk levels.

2.3.3 | Impact of perception of urgency on user convenience

Finally, we evaluated the impact of hypothetical risk on the willingness to adhere to continuous wearable use while engaging in different activities (an activity-agnostic baseline, everyday activities, physical activity, sleep, and work). The differing activities are measures of user convenience. Associations between the different activities and willingness were assessed using the Kruskal–Wallis test, with a subsequent Dunn's test for pairwise comparisons. We then individually assessed the association between the three risk levels and willingness using the Wilcoxon rank-sum test, with the different activities as stratifying variables.

TABLE 1 Breakdown of participant age, gender, age, user status, and awareness.

Demographic/group	Subgroup	Proportion (total = 359)
Age	Under 25	8.9% ($n = 32$)
	25–45	37.6% ($n = 135$)
	45–65	32.0% ($n = 115$)
	Over 65	20.3% ($n = 73$)
	Undeclared	1.1% ($n = 4$)
Gender	Man	25.6% ($n = 92$)
	Woman	70.2% ($n = 252$)
	Nonbinary/genderfluid	3.1% ($n = 11$)
	Undeclared	1.1% ($n = 4$)
User status	Users	81.9% ($n = 294$)
	Non-users	18.1% ($n = 65$)
Awareness status	High	93.6% ($n = 336$)
	Low	6.4% ($n = 23$)

3 | RESULTS

The anonymized raw survey data, as well as the analyses performed, are publicly available through dataverse: https://doi.org/10.5683/SP3/KOWQMQ https://borealisdata.ca/privateurl.xhtml?token.

3.1 | Participant demographics and status-groups

Table 1 shows the breakdown of demographics and status-groups among survey participants. The survey was engaged by 414 participants, with a completion rate of 86.7% (n = 359). Of the 359 participants, 355 provided demographic information for stratified analyses. The participants predominantly identified as female (70%) with a median age of 47 (interquartile range: 30). Overall, 82% of participants reported as wearable users (a measure of user convenience), whereas 94% reported having a high awareness of OHCA (a measure of perception of urgency).

3.2 | Design preferences impacting user convenience

In general, participants preferred hand-based form factors, with wristbands (87%), watches (86%), and rings (62%) being the most popular responses (Figure 1). In addition, comfort (94%), cost (83%), size (72%), and battery life (70%) were the four most selected factors impacting participants' decision to purchase a wearable.

3.3 | Impact of perception of urgency on the willingness to adhere to continuous wearable device use

Overall, we observed an increase in willingness to adhere with an increase in perception of urgency, represented by the increasing associated hypothetical risk of OHCA (Figures 2 and 3) ($p < 5.5 \times 10^{-14}$). For age, we found that the logistic model intercept increased, and growth/inflection parameters decreased, with increasing perceptions of urgency (Figure 3). This signifies that all ages have a high willingness to adhere at the highest perception or urgency, compared to younger individuals being significantly less willing to adhere when the perception of urgency is low (baseline risk logistic model intercept: 66.40%; growth: 0.013 year^{-1}; inflection: −53.15 year).

For the categorical demographics (gender, user status, and awareness status), we observed an increasing willingness to adhere as the perception of urgency increased across all groups (Figure 2). For gender, we did not observe significant differences in willingness to adhere between gender subgroups at the different risk levels, with the exception of the nonbinary and genderfluid (NB and GF) group at baseline risk. For user status, we found that being a wearable user (current or past user) was associated with an increased mean willingness to adhere at all risk levels (baseline risk, p-value: 0.027; IR 1, p-value: 0.0054; IR 2, p-value: 0.0097). However, we did not detect an association between having high awareness of OHCA and willingness to adhere within individual risk levels. Finally, we observed that increasing perception of urgency normalized differences in willingness to adhere between the groups (visualized most clearly for gender and awareness status).

3.4 | Impact of user convenience on perception of urgency

Overall, willingness to adhere increased with increased perceived urgency regardless of the convenience of all prompted activities (Figure 4). The participant-reported ranking of activities remained consistent across both IR levels, although the willingness to adhere ranges significantly decreased with increasing perception of urgency. For example, the largest difference in mean willingness to adhere at baseline risk was between the "sleep" (71.02%) and "physical activity" (86.26%) groups. At IR 2, the mean willingness to adhere during sleep and physical activity rose to 90.16% and 94.94%, respectively.

4 | LIMITATIONS

Participants were predominantly located in British Columbia, Canada and recruited through ReachBC. Such platforms tend to attract individuals involved in prior research studies who may have greater knowledge of healthcare attributes compared to the general population. This would indicate that observed trends represent individuals

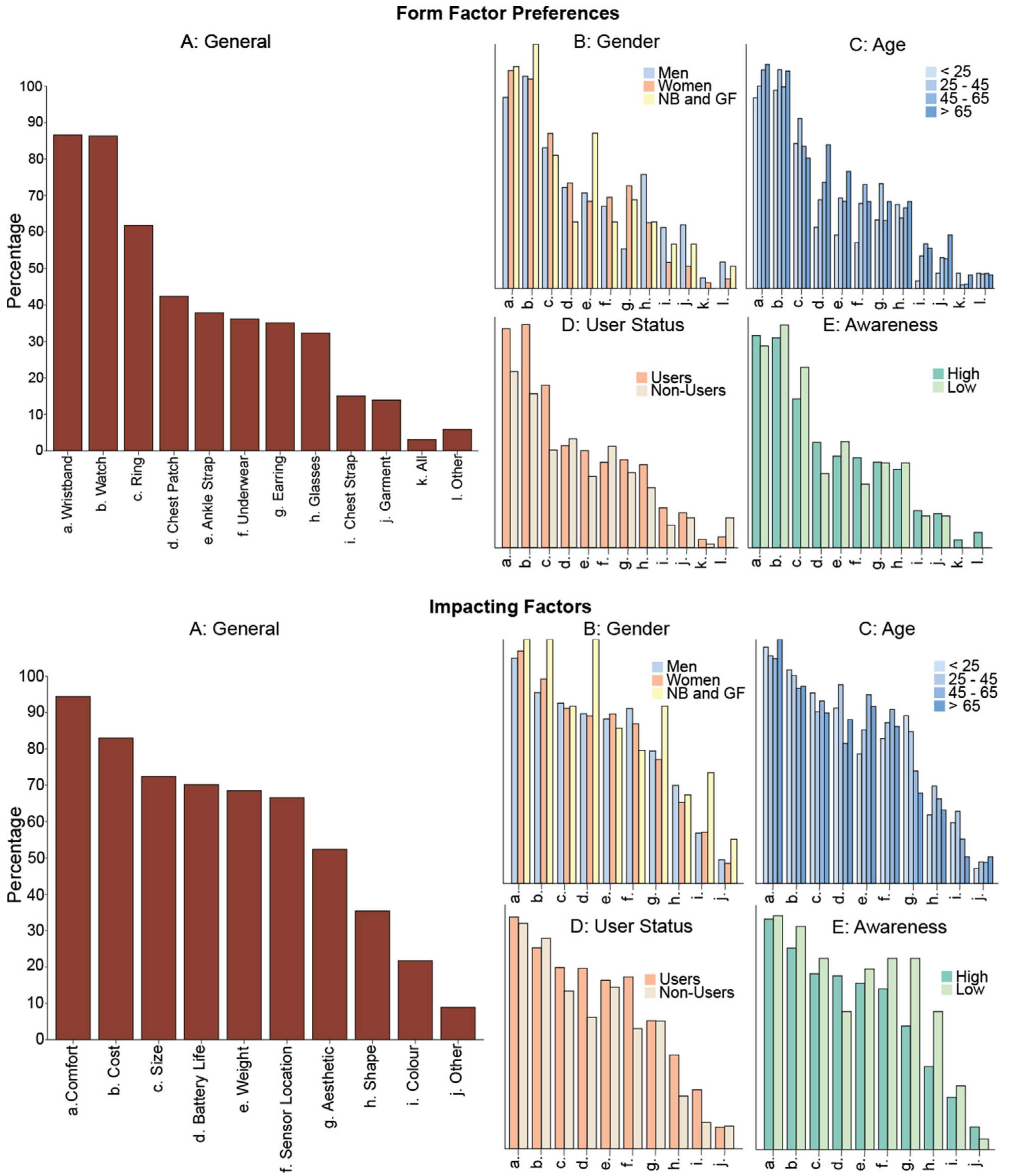

FIGURE 1 Form factor and usability preferences (A) by gender (B), age (C), user status (D), and awareness (E). All plots have identical axes. Abbreviations: NB and GF: nonbinary and genderfluid.

who would generally be more knowledgeable of cardiac conditions and their consequences. Additionally, the survey topic was included in all outreach, potentially introducing selection bias. As such, the trends observed in other populations might reflect a lower willingness to adhere, further highlighting the importance of awareness surrounding OHCA. In addition, the reported measure in this survey is hypothetical willingness to adhere to continuous wearable use, which may differ from actual adherence once such wearables are deployed. The suggested hypothetical device is also presented as having 100% accuracy, neglecting false positives and potentially increasing participants' willingness to adhere. Thus, these results represent an upper bound on the willingness to use a perfect sensor, and additional factors

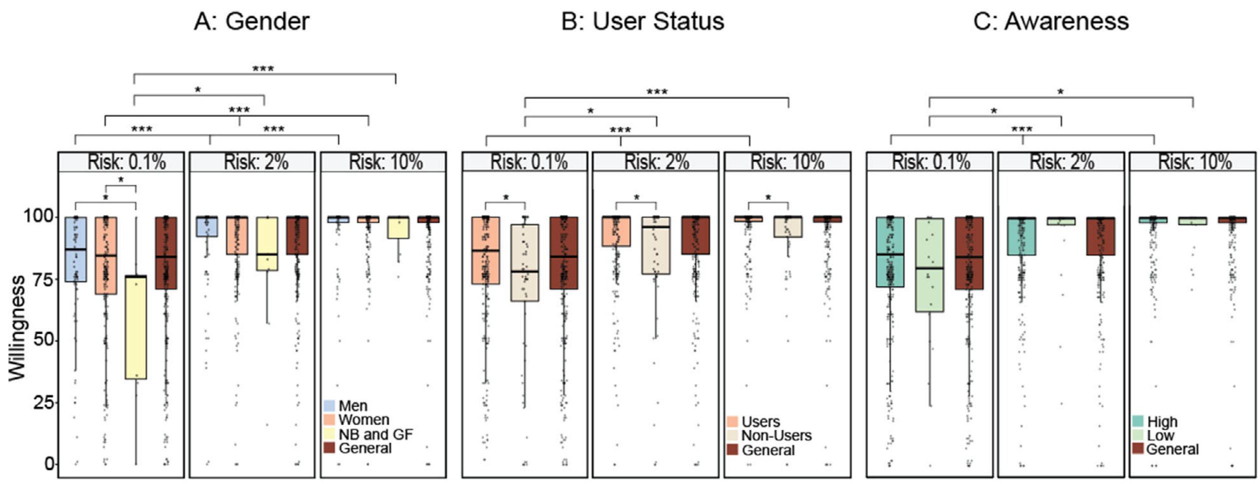

FIGURE 2 Impact of hypothetical risk on willingness to continuously use a wearable for out-of-hospital cardiac arrest (OHCA) by gender (A), user status (B), and awareness (C). General: All participants. Significance levels: ***$p \leq 0.001$; *$p \leq 0.05$.

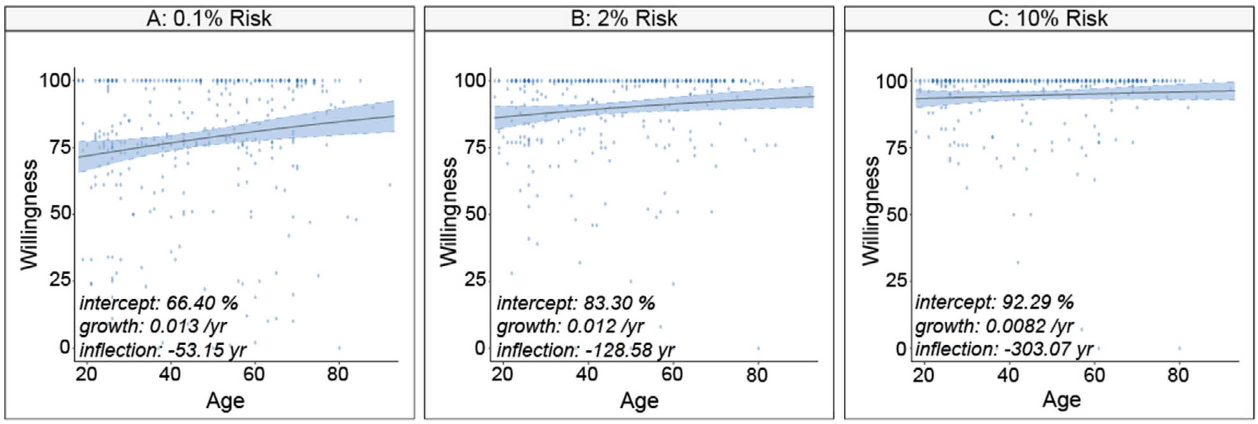

FIGURE 3 Logistic models of willingness to continuously use a wearable for out-of-hospital cardiac arrest (OHCA) as a function of age at baseline risk (A), IR 1 (B), and IR 2 (C).

such as false positive rates will be further investigated in future work.

Despite a large number of complete entries (355), the NB and GF, male-identifying, and low awareness groups contained small sample sizes ($n = 11$, 92, and 23, respectively). This may have affected observed trends. In addition, awareness was based solely on personal/family medical history and personally knowing someone who has experienced cardiac arrest, which may not fully encompass the participants' level of awareness surrounding OHCA.

5 | DISCUSSION

5.1 | Summary

We deployed a survey to investigate the user convenience and user perception of urgency factors associated with participant-reported willingness to adhere to continuous wearable use. Our results suggest that there are certain factors that influence a user's willingness to adhere. First, most users prefer hand-based devices (watches, wristbands, and rings) that are comfortable, cheap, small, and have a long battery life. Second, increasing the hypothetical risk of OHCA dramatically increased willingness to adhere and normalized demographic trends (eg, older individuals were more willing to adhere at baseline risk, but all age groups had high willingness at IR 2). This suggests that adherence may be improved using a multifactored approach by both improving the engineering design of wearable devices (user convenience) and potentially improving education on the risks of OHCA (perception of urgency).

5.2 | Perception of urgency and willingness to adhere to continuous wearable use

At the baseline risk of 0.1%, the group most willing to adhere is as expected: older individuals with a history of cardiovascular health

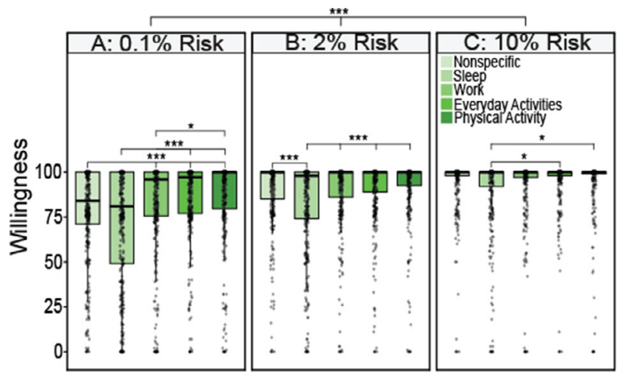

FIGURE 4 Impact of hypothetical risk on willingness to continuously use a wearable for out-of-hospital cardiac arrest (OHCA) during various prompted activities at baseline risk (A), IR 1 (B), and IR 2 (C). Significance levels: ***$p \leq 0.001$; *$p \leq 0.05$.

issues who are familiar with wearables. However, as hypothetical risk increased, differences in willingness across demographic and group status decreased. This is most evident in the logistic model of willingness as a function of age. The intercept and slope of the model at IR 2 indicate that, when individuals perceive their risk of experiencing an OHCA to be high, willingness to adhere lies above 90% for all age groups.

However, at baseline 0.1% risk, the perception of urgency is lower. There is a distinction between risk and perceived urgency. Perceived urgency may be high even when the risk is low, because some disease states have tremendous consequences for an individual's well-being. However, this is not what we observed from our survey in the context of OHCA.

A 2017 survey from Cleveland Clinic revealed that 87% of Americans use "cardiac arrest" and "heart attack" interchangeably.[28] While a "heart attack" is a serious cardiac issue, it does not result in immediate loss of consciousness and has a much higher survival rate (90%–97%).[29] Pairing a 0.1% risk of OHCA with a perceived 90%–97% survival could account for a low perception of urgency.

Participants indicated the highest willingness to adhere while engaging in physical activity and the lowest while asleep. However, OHCAs occurring during the nighttime have a significantly lower survival rate (odds ratio: 1.20–1.66)[30,31] and account for 14.6% of all sudden cardiac deaths.[32] This represents a discrepancy between participant preferences and optimal OHCA detection periods. Participants may perceive wearable use during sleep to be uncomfortable and may be most willing to use a wearable during physical activity due to the prevalence of commercial devices focused on athletic performance.

5.3 | User convenience and willingness to adhere to continuous wearable use

A recent publication by Hutton et al. emphasizes the importance of recognition on survival from OHCA to hospital discharge, suggesting a 147% increase in survival if unwitnessed cases were detected by a biosensor.[6] A key element in the success of these wearables is continuous use to ensure that if an OHCA were to occur, the device will be able to detect it. If 87% or 86% of users would wear a wristband or wristwatch, respectively, as indicated by our survey, then Hutton et al. suggest survival to hospital discharge due to continuous use of a wearable could increase by 136%. For rings (62%), this could result in a 90% increase.

While the current state of wearables favors hand-based devices (eg, wristwatches and rings), our results suggest that comfort (94%) and size (72%) are the driving factors influencing preferences. Future form factors not addressed in our study may be favorable if they adequately address these factors.

In conclusion, our findings suggest that a majority of participants would be willing to continuously use wearables to monitor for OHCA. Furthermore, we suggest ways in which willingness to adhere may improve, focused on increasing perception of urgency. Groups with increased perception of urgency were more willing to adhere to wearable use regardless of age, gender, and user status. These findings highlight the factors related to the acceptability of wearing OHCA sensors and the potential importance of higher awareness to ensure the adoption of an OHCA-specific wearable in the general population.

AUTHOR CONTRIBUTIONS

Saud Lingawi is the primary author responsible for conceptualization, data collection and analysis, critical evaluation of the results, and manuscript preparation. Jacob Hutton assisted in the development of statistical analysis methods. Calvin Kuo is the senior author responsible for editing and supervision. Mahsa Khalili is the MSHRBC researcher, and Brian Grunau is the recipient of MITACS. Remaining authors provided editing suggestions.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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